Substrate Framework: Falsifiable Observational Predictions

Eliot Gable*
Independent Researcher
(Dated: 2026-05-05)

A live audit of the substrate-framework's quantitative predictions across particle, gravitational, and cosmological observables. Each predicted value is paired with the current best measurement and the deviation, anchored on three independent reading paths (CODATA m_e, CODATA m_μ [3], and the framework's own Higgs vev ≈ 246.21965 GeV). All values computed at mp.prec = 500. Status flags: † tension, ‡ open prediction, †† regime-tension. As of 2026-05-05, no entry is in falsifying tension ($\geq 3\sigma$) with observation. Three entries (ρ_Λ, H_0 tension magnitude, TOV bound) are flagged ‡ pending closure of upstream prediction work; 38 distinct observables are auto-flagged blue (estimate-pending); 14 half-integer-slot dark-mode candidates are auto-flagged violet (mass predicted, but the framework cannot yet decide whether the slot is occupied). Cell counts in Tables 1–2 exceed observable counts because each observable is rendered across three anchor columns. This document is intended as a falsification target.

I. METHODOLOGY

Predictions are computed from a single mass-anchor parameter h_{eff} that is independently fixed three ways: from CODATA m_e [3], from CODATA m_μ, and from the framework's own Higgs vev. The three anchor reading paths agree to $\sim$ 158 ppt (max pairwise relative deviation across the full 42-particle catalogue), with the spread carried mostly by the e-anchor-vs-vev-anchor pair ($\sim$ 158 ppt) and the μ-anchor-vs-vev-anchor pair ($\sim$ 153 ppt); the e-anchor-vs-μ-anchor pair agrees at $\sim$ 4 ppt. All computations are performed at mp.prec = 500 in mpmath; no floating-point compounding occurs.

a. What each anchor column means. The three anchor columns in Tables 1 and 2 are not three independent measurements; each carries a specific epistemic status:

- The **electron-anchor** column inherits the electron-mass CODATA central value as its base reading; every prediction in that column is propagated from that single experimental input. The column's residual against any other observable is therefore bounded below by the CODATA m_e experimental uncertainty (and above by the framework's structural precision).
- The **muon-anchor** column does the same with the muon-mass CODATA central value as its single experimental input.
- The **vev-anchor** column is qualitatively different: it carries the framework's geometry-intrinsic prediction of the Higgs vev as its base reading, with no CODATA particle-mass input. The values in that column are pure framework predictions; their residual against observation is not bounded below by any experimental uncertainty band on a measured anchor.

A reader comparing the three columns is therefore comparing two experimentally-anchored readings against one geometry-only reading; agreement of all three to $\sim$ 4 ppt is consistency between two independent CODATA inputs and one parameter-free geometric prediction.

b. Auto-flag legend (appears on every cell). Each predicted value's deviation against the observed value is automatically compared against the observation's published experimental uncertainty σ_{obs} (covering $\sim$ 70 observables):

- ✓ **green:** $|\text{dev}| \leq 1\,\sigma_{\text{obs}}$ — prediction sits within the experimental band.
- • **orange:** $1\,\sigma_{\text{obs}} < |\text{dev}| \leq 3\,\sigma_{\text{obs}}$ — marginal.
- × **red:** $|\text{dev}| > 3\,\sigma_{\text{obs}}$ — outside the experimental band; falsifying tension if reproducible.
- ? **blue:** estimate-pending; the prediction depends on upstream work that is not yet closed, so the deviation is *not yet falsifiable*. The work-remaining tally (§ Work remaining) names which item gates each blue prediction.
- ? **violet:** mass predicted, but the framework cannot yet decide whether this candidate slot is physically occupied. Used for spin-$\frac{1}{2}$ half-integer-slot dark-mode candidates.
- plain (no colour): σ_{obs} unavailable for this observable; status undetermined.

II. SM MASS SPECTRUM, ELECTROWEAK SECTOR, MIXING

* 10015996+egable@users.noreply.github.com

TABLE I: Substrate-framework predictions for all derivable Standard Model and gravitational observables, computed at mp.prec = 500 from three independent anchor scales: CODATA electron mass, CODATA muon mass, and the directly-computed structural Higgs vev $v_C = (\dim -2)/(1-\delta_W)\cdot(1+5\alpha_\tau^3/8)$. Each cell carries the predicted value (15-digit precision) with the deviation against the observed value beneath, in the same cell. Auto-flag legend: ✓ green ($|\text{dev}| \le 1\sigma_{\text{obs}}$), • orange ($1\sigma < |\text{dev}| \le 3\sigma$), × red ($|\text{dev}| > 3\sigma$), ? blue (estimate-pending; the prediction depends on upstream prediction work that is not yet closed, so the deviation is not yet falsifiable), ? violet (mass predicted, but the framework cannot yet decide whether this candidate slot is physically occupied; used for half-integer-slot dark-mode candidates). Observed column carries the published experimental uncertainty (± ppm/ppb/ppt) below each value where available. Anchor variance (max pairwise ∼ 158 ppt between anchors across the full 42-particle catalogue, carried by the e-anchor-vs-vev-anchor pair; e-anchor-vs-μ-anchor agrees at ∼ 4 ppt) is exposed rather than hidden.

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
CHARGED LEPTON MASSES				
electron (MeV)	0.5109989500000 ±300.000 ppt	0.5109989500000 ✓ ∼ 0 (FP floor)	0.5109989500022 ✓ + 4.255 ppt	0.5109989500806 ✓ + 157.642 ppt
muon (MeV)	105.6583755000 ±22.000 ppb	105.6583754996 ✓ − 4.255 ppt	105.6583755000 ✓ ∼ 0 (FP floor)	105.6583755162 ✓ + 153.386 ppt
tau (MeV)	1776.860000000 ±68.000 ppm	1776.859738841 ✓ − 146.978 ppb	1776.859738849 ✓ − 146.973 ppb	1776.859739121 ✓ − 146.820 ppb
QUARK MASSES				
up (MeV)	2.160000000000 ±170000.000 ppm	2.172135018407 ✓ + 5618.064 ppm	2.172135018416 ✓ + 5618.064 ppm	2.172135018749 ✓ + 5618.064 ppm
down (MeV)	4.670000000000 ±70000.000 ppm	4.666472667699 ✓ − 755.317 ppm	4.666472667719 ✓ − 755.317 ppm	4.666472668435 ✓ − 755.317 ppm
strange (MeV)	93.40000000000 ±92000.000 ppm	93.25272875250 ✓ − 1576.780 ppm	93.25272875289 ✓ − 1576.780 ppm	93.25272876720 ✓ − 1576.780 ppm
charm (MeV)	1273.000000000 ±3600.000 ppm	1274.297660596 ✓ + 1019.372 ppm	1274.297660601 ✓ + 1019.372 ppm	1274.297660796 ✓ + 1019.372 ppm
bottom (MeV)	4183.000000000 ±1700.000 ppm	4178.924486322 ✓ − 974.304 ppm	4178.924486340 ✓ − 974.304 ppm	4178.924486981 ✓ − 974.304 ppm
top (GeV)	172.5700000000 ±1700.000 ppm	172.5698832743 ✓ − 676.396 ppb	172.5698832751 ✓ − 676.392 ppb	172.5698833015 ✓ − 676.238 ppb
NEUTRINO MASSES				
neutrino 1 (meV)	4.510000000000e-10 ±50000.000 ppm	4.516090890957e-10 ✓ + 1350.530 ppm	4.516090890976e-10 ✓ + 1350.530 ppm	4.516090891669e-10 ✓ + 1350.530 ppm
neutrino 2 (meV)	8.613900000000e-9 ±50000.000 ppm	8.625379528706e-9 ✓ + 1332.675 ppm	8.625379528743e-9 ✓ + 1332.675 ppm	8.625379530066e-9 ✓ + 1332.675 ppm
neutrino 3 (meV)	5.014980000000e-8 ±50000.000 ppm	5.021285707439e-8 ✓ + 1257.374 ppm	5.021285707461e-8 ✓ + 1257.374 ppm	5.021285708231e-8 ✓ + 1257.375 ppm
ELECTROWEAK SECTOR				
Higgs vev (GeV)	246.2196500000 ±24.000 ppb	246.2196499809 ✓ − 77.617 ppt	246.2196499819 ✓ − 73.361 ppt	246.2196500197 ✓ + 80.025 ppt
Higgs boson (GeV)	125.2000000000 ±880.000 ppm	125.1997813344 ✓ − 1.747 ppm	125.1997813350 ✓ − 1.747 ppm	125.1997813542 ✓ − 1.746 ppm
W boson (GeV)	80.36920000000 ±160.000 ppm	80.36920037081 ✓ + 4.614 ppb	80.36920037115 ✓ + 4.618 ppb	80.36920038347 ✓ + 4.771 ppb
Z boson (GeV)	91.18760000000 ±23.000 ppm	91.19003480815 ? + 26.701 ppm	91.19003480853 ? + 26.701 ppm	91.19003482252 ? + 26.701 ppm
Weinberg mixing fraction	0.2232000000000 ±140.000 ppm	0.2232441517887 ? + 197.813 ppm	0.2232441517887 ? + 197.813 ppm	0.2232441517887 ? + 197.813 ppm
SU(2) coupling	0.6533000000000 ±1500.000 ppm	0.6528252345176 ? − 726.719 ppm	0.6528252345176 ? − 726.719 ppm	0.6528252345176 ? − 726.719 ppm
$U(1)_Y$ coupling (EW norm)	0.3500000000000 ±1000.000 ppm	0.3499811975896 ? − 53.721 ppm	0.3499811975896 ? − 53.721 ppm	0.3499811975896 ? − 53.721 ppm
$U(1)_Y$ coupling (GUT norm)	0.4519000000000 ±1000.000 ppm	0.4518237832501 ? − 168.658 ppm	0.4518237832501 ? − 168.658 ppm	0.4518237832501 ? − 168.658 ppm
SU(2) fine-structure	0.03394000000000 ±1500.000 ppm	0.03391438943683 ? − 754.583 ppm	0.03391438943683 ? − 754.583 ppm	0.03391438943683 ? − 754.583 ppm
$U(1)_Y$ fine-structure (EW)	0.009745000000000 ±2000.000 ppm	0.009747192918717 ? + 225.030 ppm	0.009747192918717 ? + 225.030 ppm	0.009747192918717 ? + 225.030 ppm
$U(1)_Y$ fine-structure (GUT)	0.01624100000000 ±2000.000 ppm	0.01624532153119 ? + 266.088 ppm	0.01624532153119 ? + 266.088 ppm	0.01624532153119 ? + 266.088 ppm
EM fine-structure (tree)	0.007575408000000 ±1.000 ppb	0.007571189103257 ? − 556.920 ppm	0.007571189103257 ? − 556.920 ppm	0.007571189103257 ? − 556.920 ppm
LEPTON YUKAWA COUPLINGS				
electron Yukawa	2.935028319017e-6 ±300.000 ppt	2.935028319245e-6 ✓ + 77.617 ppt	2.935028319245e-6 ✓ + 77.617 ppt	2.935028319245e-6 ✓ + 77.617 ppt
muon Yukawa	0.0006068707660433 ±22.000 ppb	0.0006068707660878 ✓ + 73.361 ppt	0.0006068707660878 ✓ + 73.361 ppt	0.0006068707660878 ✓ + 73.361 ppt
tau Yukawa	0.01020576347354 ±68.000 ppm	0.01020576197431 ✓ − 146.900 ppb	0.01020576197431 ✓ − 146.900 ppb	0.01020576197431 ✓ − 146.900 ppb
QUARK YUKAWA COUPLINGS				

(continued on next page)

Table I continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
up Yukawa	1.240640742819e-5 ±170000.000 ppm	1.247610742106e-5 ✓ + 5618.064 ppm	1.247610742106e-5 ✓ + 5618.064 ppm	1.247610742106e-5 ✓ + 5618.064 ppm
down Yukawa	2.682311235631e-5 ±70000.000 ppm	2.680285239466e-5 ✓ − 755.317 ppm	2.680285239466e-5 ✓ − 755.317 ppm	2.680285239466e-5 ✓ − 755.317 ppm
strange Yukawa	0.0005364622471263 ±92000.000 ppm	0.0005356163642517 ✓ − 1576.780 ppm	0.0005356163642517 ✓ − 1576.780 ppm	0.0005356163642517 ✓ − 1576.780 ppm
charm Yukawa	0.007311739192631 ±3600.000 ppm	0.007319192575631 ✓ + 1019.372 ppm	0.007319192575631 ✓ + 1019.372 ppm	0.007319192575631 ✓ + 1019.372 ppm
bottom Yukawa	0.02402592697783 ±1700.000 ppm	0.02400251842267 ✓ − 974.304 ppm	0.02400251842267 ✓ − 974.304 ppm	0.02400251842267 ✓ − 974.304 ppm
top Yukawa	0.9911915416122 ±1700.000 ppm	0.9911908712511 ✓ − 676.318 ppb	0.9911908712511 ✓ − 676.318 ppb	0.9911908712511 ✓ − 676.318 ppb
HIGGS SELF-COUPLING				
Higgs self-coupling	0.1292805654113 ±1800.000 ppm	0.1292801138469 ✓ − 3.493 ppm	0.1292801138469 ✓ − 3.493 ppm	0.1292801138469 ✓ − 3.493 ppm
STRONG COUPLING				
strong coupling	0.1180000000000 ±7600.000 ppm	0.1180478309232 ? + 405.347 ppm	0.1180478309232 ? + 405.347 ppm	0.1180478309232 ? + 405.347 ppm
PMNS NEUTRINO MIXING				
PMNS solar mixing fraction	0.3030000000000 ±50000.000 ppm	0.3065788699420 ? + 11811.452 ppm	0.3065788699420 ? + 11811.452 ppm	0.3065788699420 ? + 11811.452 ppm
PMNS atmospheric mixing fraction	0.5720000000000 ±30000.000 ppm	0.5714285714286 ? − 999.001 ppm	0.5714285714286 ? − 999.001 ppm	0.5714285714286 ? − 999.001 ppm
PMNS reactor mixing fraction	0.02203000000000 ±30000.000 ppm	0.02197802197802 ? − 2359.420 ppm	0.02197802197802 ? − 2359.420 ppm	0.02197802197802 ? − 2359.420 ppm
PMNS CP phase (rad)	3.438298626429 ±50000.000 ppm	3.455751918949 ? + 5076.142 ppm	3.455751918949 ? + 5076.142 ppm	3.455751918949 ? + 5076.142 ppm
CKM QUARK MIXING				
Cabibbo mixing fraction	0.05064000000000 ±4000.000 ppm	0.05035971223022 ? − 5534.909 ppm	0.05035971223022 ? − 5534.909 ppm	0.05035971223022 ? − 5534.909 ppm
CKM atmospheric mixing fraction	0.001682000000000 ±15000.000 ppm	0.001681796794019 ? − 120.812 ppm	0.001681796794019 ? − 120.812 ppm	0.001681796794019 ? − 120.812 ppm
CKM reactor mixing fraction	1.460000000000e-5 ±30000.000 ppm	1.452605387890e-5 ? − 5064.803 ppm	1.452605387890e-5 ? − 5064.803 ppm	1.452605387890e-5 ? − 5064.803 ppm
CKM CP phase (rad)	1.143190660056 ±50000.000 ppm	1.151917306316 ? + 7633.588 ppm	1.151917306316 ? + 7633.588 ppm	1.151917306316 ? + 7633.588 ppm
QCD THETA ANGLE				
QCD theta angle (rad)	0.0	0.0	0.0	0.0
MASSLESS GAUGE BOSONS				
photon (MeV)	0.0	0.0	0.0	0.0
gluon 1 (MeV)	0.0	0.0	0.0	0.0
gluon 2 (MeV)	0.0	0.0	0.0	0.0
gluon 3 (MeV)	0.0	0.0	0.0	0.0
gluon 4 (MeV)	0.0	0.0	0.0	0.0
gluon 5 (MeV)	0.0	0.0	0.0	0.0
gluon 6 (MeV)	0.0	0.0	0.0	0.0
gluon 7 (MeV)	0.0	0.0	0.0	0.0
gluon 8 (MeV)	0.0	0.0	0.0	0.0
GRAVITATIONAL SECTOR				
Newton's G ($m^3kg^{-1}s^{-2}$)	6.674300000000e-11 ±22.000 ppm	6.674316910358e-11 ✓ + 2.534 ppm	6.674316910301e-11 ✓ + 2.534 ppm	6.674316908253e-11 ✓ + 2.533 ppm
Planck mass (GeV)	1.220890000000e+22 ±11.000 ppm	1.220888581931e+22 ✓ − 1.162 ppm	1.220888581936e+22 ✓ − 1.161 ppm	1.220888582124e+22 ✓ − 1.161 ppm
ANOMALOUS MAGNETIC MOMENTS				
electron anomalous moment	0.001159652181000 ±1.700 ppt	0.001160667181068 ? + 875.263 ppm	0.001160667181068 ? + 875.263 ppm	0.001160667181068 ? + 875.263 ppm
muon anomalous moment	0.001165920620000 ±460.000 ppb	0.001166944225438 ? + 877.938 ppm	0.001166944225438 ? + 877.938 ppm	0.001166944225438 ? + 877.938 ppm
tau anomalous moment	0.001177210000000 ±2100.000 ppm	0.001176416612358 ? − 673.956 ppm	0.001176416612358 ? − 673.956 ppm	0.001176416612358 ? − 673.956 ppm
FERMI CONSTANT				
Fermi constant (GeV^{-2})	1.166378700000e-5 ±500.000 ppb	1.166378707705e-5 ✓ + 6.606 ppb	1.166378707695e-5 ✓ + 6.597 ppb	1.166378707337e-5 ✓ + 6.291 ppb
LEPTON LIFETIMES				
muon lifetime (s)	2.196981100000e-6 ±1.000 ppm	2.197000758647e-6 ? + 8.948 ppm	2.197000758638e-6 ? + 8.948 ppm	2.197000758301e-6 ? + 8.948 ppm
tau lifetime (s)	2.903000000000e-13 ±400.000 ppm	2.975066653724e-13 ? + 24824.889 ppm	2.975066653712e-13 ? + 24824.889 ppm	2.975066653255e-13 ? + 24824.889 ppm

(continued on next page)

Table I continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
W AND Z DECAY WIDTHS				
W total decay width (GeV)	2.085000000000 ±2000.000 ppm	2.097035187050 ? + 5772.272 ppm	2.097035187059 ? + 5772.272 ppm	2.097035187381 ? + 5772.272 ppm
Z total decay width (GeV)	2.495500000000 ±90.000 ppm	2.508962908085 ? + 5394.874 ppm	2.508962908096 ? + 5394.874 ppm	2.508962908481 ? + 5394.874 ppm
Z invisible width (3 neutrinos) (GeV)	0.4995000000000 ±3000.000 ppm	0.4976876518476 ? − 3628.325 ppm	0.4976876518498 ? − 3628.325 ppm	0.4976876519261 ? − 3628.324 ppm
Z leptonic width (per lepton) (GeV)	0.08398000000000 ±500.000 ppm	0.08389802825078 ? − 976.087 ppm	0.08389802825113 ? − 976.087 ppm	0.08389802826400 ? − 976.086 ppm
HIGGS DECAY WIDTHS				
Higgs width to bb (GeV)	0.002200000000000 ±140000.000 ppm	0.002303033296527 ? + 46833.317 ppm	0.002303033296517 ? + 46833.317 ppm	0.002303033296166 ? + 46833.316 ppm
Higgs width to tau tau (GeV)	0.0002300000000000 ±100000.000 ppm	0.0002591191584134 ? + 126605.037 ppm	0.0002591191584145 ? + 126605.037 ppm	0.0002591191584543 ? + 126605.037 ppm
Higgs width to cc (GeV)	0.0001200000000000 ±300000.000 ppm	0.0001140529647702 ? − 49558.627 ppm	0.0001140529647697 ? − 49558.627 ppm	0.0001140529647522 ? − 49558.627 ppm
Higgs total fermionic width (GeV)	0.002510000000000 ±50000.000 ppm	0.002676205419710 ? + 66217.299 ppm	0.002676205419701 ? + 66217.299 ppm	0.002676205419372 ? + 66217.299 ppm
COMPTON WAVELENGTHS AND ATOMIC CONSTANTS				
electron Compton wavelength (fm)	2426.310238670 ±1.500 ppb	2426.310237936 ✓ − 302.639 ppt	2426.310237925 ✓ − 306.895 ppt	2426.310237553 ✓ − 460.281 ppt
muon Compton wavelength (fm)	11.73444110000 ±22.000 ppb	11.73444109942 ✓ − 49.815 ppt	11.73444109937 ✓ − 54.070 ppt	11.73444109757 ✓ − 207.457 ppt
tau Compton wavelength (fm)	0.6977713400000 ±70.000 ppm	0.6977714429885 ✓ + 147.596 ppb	0.6977714429855 ✓ + 147.592 ppb	0.6977714428785 ✓ + 147.439 ppb
Bohr radius (fm)	52917.72109000 ±1.500 ppb	52871.37492741 ? − 875.816 ppm	52871.37492719 ? − 875.816 ppm	52871.37491908 ? − 875.816 ppm
Rydberg energy (eV)	13.60569312300 ±14.000 ppt	13.62955661536 ? + 1753.934 ppm	13.62955661542 ? + 1753.934 ppm	13.62955661751 ? + 1753.934 ppm
EFFECTIVE NEUTRINO SPECIES				
effective neutrino species	3.046000000000 ±56000.000 ppm	3.000000000000 ? − 15101.773 ppm	3.000000000000 ? − 15101.773 ppm	3.000000000000 ? − 15101.773 ppm
DARK MATTER ULTRALIGHT BAND (predicted-but-unobserved; falsification targets)				
DM ultralight 1 (heaviest) (eV)	–	6.205011597344e-7 ? predicted	6.205011597370e-7 ? predicted	6.205011598322e-7 ? predicted
DM ultralight 2 (eV)	–	2.232018560196e-9 ? predicted	2.232018560205e-9 ? predicted	2.232018560548e-9 ? predicted
DM ultralight 3 (eV)	–	8.028843741711e-12 ? predicted	8.028843741745e-12 ? predicted	8.028843742977e-12 ? predicted
DM ultralight 4 (eV)	–	2.888073288385e-14 ? predicted	2.888073288398e-14 ? predicted	2.888073288841e-14 ? predicted
DM ultralight 5 (eV)	–	1.038875283592e-16 ? predicted	1.038875283596e-16 ? predicted	1.038875283756e-16 ? predicted
DM ultralight 6 (eV)	–	3.736961451769e-19 ? predicted	3.736961451785e-19 ? predicted	3.736961452358e-19 ? predicted
DM ultralight 7 (lightest) (eV)	–	1.344230738046e-21 ? predicted	1.344230738052e-21 ? predicted	1.344230738258e-21 ? predicted
DARK MATTER MeV–GeV BAND (predicted-but-unobserved; direct-detection targets)				
DM MeV–GeV mode A (MeV)	–	0.8335357316677 ? predicted	0.8335357316712 ? predicted	0.8335357317991 ? predicted
DM MeV–GeV mode B (MeV)	–	5.828295547031 ? predicted	5.828295547056 ? predicted	5.828295547950 ? predicted
DM MeV–GeV mode C (MeV)	–	46.62636437625 ? predicted	46.62636437645 ? predicted	46.62636438360 ? predicted
DM MeV–GeV mode D (GeV)	–	1.492043660040 ? predicted	1.492043660046 ? predicted	1.492043660275 ? predicted
DM MeV–GeV mode E (GeV)	–	4.816169313316 ? predicted	4.816169313336 ? predicted	4.816169314075 ? predicted
DM MeV–GeV mode F (GeV)	–	592.3496671013 ? predicted	592.3496671038 ? predicted	592.3496671946 ? predicted

(continued on next page)

Table I continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
HALF-SLOT CANDIDATES (spin-1/2 dark-mode candidates; slot occupancy open)				
DM half-slot candidate 1 (MeV)	–	0.6552852638606 ? slot open	0.6552852638634 ? slot open	0.6552852639639 ? slot open
DM half-slot candidate 2 (MeV)	–	2.775832922367 ? slot open	2.775832922379 ? slot open	2.775832922805 ? slot open
DM half-slot candidate 3 (MeV)	–	7.267224937849 ? slot open	7.267224937879 ? slot open	7.267224938994 ? slot open
DM half-slot candidate 4 (MeV)	–	11.75861695333 ? slot open	11.75861695338 ? slot open	11.75861695518 ? slot open
DM half-slot candidate 5 (MeV)	–	30.78445884451 ? slot open	30.78445884464 ? slot open	30.78445884936 ? slot open
DM half-slot candidate 6 (MeV)	–	49.81030073569 ? slot open	49.81030073590 ? slot open	49.81030074354 ? slot open
DM half-slot candidate 7 (MeV)	–	130.4050603159 ? slot open	130.4050603164 ? slot open	130.4050603364 ? slot open
DM half-slot candidate 8 (MeV)	–	552.4047001080 ? slot open	552.4047001104 ? slot open	552.4047001951 ? slot open
DM half-slot candidate 9 (MeV)	–	0.6126241040672 ? slot open	0.6126241040698 ? slot open	0.6126241041638 ? slot open
DM half-slot candidate 10 (MeV)	–	1.732762633217 ? slot open	1.732762633224 ? slot open	1.732762633490 ? slot open
DM half-slot candidate 11 (MeV)	–	3.465525266434 ? slot open	3.465525266449 ? slot open	3.465525266980 ? slot open
DM half-slot candidate 12 (MeV)	–	9.801985665075 ? slot open	9.801985665117 ? slot open	9.801985666620 ? slot open
DM half-slot candidate 13 (MeV)	–	13.86210106574 ? slot open	13.86210106580 ? slot open	13.86210106792 ? slot open
DM half-slot candidate 14 (MeV)	–	27.72420213147 ? slot open	27.72420213159 ? slot open	27.72420213584 ? slot open
RULED OUT BY FRAMEWORK				
4th chiral generation	**ruled out:** Framework predicts exactly three chiral fermion generations. **Falsified** if a 4th chiral lepton or quark generation is observed.			
Fundamental scalar beyond Higgs	**ruled out:** Framework predicts the Higgs is the unique fundamental scalar. **Falsified** if a non-Higgs fundamental scalar is observed.			
Quantum-gravity scale below $\sim 10\,E_{\rm Planck}$	**ruled out:** Framework predicts no quantum-gravity scale below $\sim 10\,E_{\rm Planck}$. **Falsified** if quantum-gravity signatures appear at sub-Planckian energies.			
Ultralight DM modes below $\sim 10^{-21}$ eV	**ruled out:** Framework predicts no ultralight-DM modes below $\sim 10^{-21}$ eV (Lyman-α structure-formation cutoff, Rogers & Peiris 2021). **Falsified** if direct detection finds a discrete-spectrum DM mode in this range.			
PQ axion (strong CP)	**ruled out:** Framework predicts $\theta_{\rm QCD} = 0$ exactly with no PQ axion required. **Falsified** if $\theta_{\rm QCD}$ is measured nonzero.			
Dark photon (kinetic-mixing operator)	**ruled out:** Framework predicts no 4D kinetic-mixing dark-photon operator. **Falsified** if dark-photon kinetic-mixing-style coupling is detected.			

III. GRAVITATIONAL AND COSMOLOGICAL OBSERVABLES

TABLE II: Extended framework predictions: gravitational sector, cosmological observables, and additional constants. **Honest scope:** the framework's genuine gravity-sector predictions are exactly two — Newton's G ($\sim$ 2500 ppb) and the Planck mass m_P ($\sim$ 1 ppb). All other gravity-sector entries (Schwarzschild radii, perihelion precessions, light deflection, Shapiro delay, gravitational redshift) are GR consequences computed using the framework's predicted G together with *observed* body parameters ($M_\odot$, $M_\oplus$, $R_\odot$, etc.); they are GR-consistency checks of the framework G, not independent predictions. The MOND a_0 and $\Lambda_{\mathrm{crit-tension}}$ rows are literature MOND (Milgrom 1983; Famaey & Binney 2005) treated as ansatz; closure of the underlying equations is open. Computed at mp.prec = 500, anchored on three independent reading paths (CODATA m_e, CODATA m_μ, framework Higgs vev). Each predicted value is shown to 15-digit precision with the deviation against the observed value beneath. The Observed column carries the published experimental uncertainty (± ppm/ppb/ppt) below each value where available. † tension; ‡ open prediction; †† regime-tension; ? pending closure (literature ansatz or GR-consistency check).

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
Higgs vev (structural identity) (GeV)	246.2196500000 ±24.368 ppm	246.2196500197 ✓ + 80.025 ppt	246.2196500197 ✓ + 80.025 ppt	246.2196500197 ✓ + 80.025 ppt
Newton's G ($\mathrm{m^3kg^{-1}s^{-2}}$)	6.674300000000e-11 ±22.474 ppm	6.674316918458e-11 ✓ + 2.535 ppm	6.674316918401e-11 ✓ + 2.535 ppm	6.674316916354e-11 ✓ + 2.535 ppm
Planck mass m_P (GeV)	1.220890000000e+19 ±11.467 ppm	1.220888581931e+19 ✓ − 1.162 ppm	1.220888581936e+19 ✓ − 1.161 ppm	1.220888582124e+19 ✓ − 1.161 ppm
GR perihelion precession (Mercury) (arcsec/century)	43.11000000000 ±4871.260 ppm	42.99184861659 ✓ − 2740.696 ppm	42.99184861659 ✓ − 2740.696 ppm	42.99184861659 ✓ − 2740.696 ppm
GR perihelion precession (Venus) (arcsec/century)	8.626000000000 ±3477.858 ppm	8.626741339097 ✓ + 85.942 ppm	8.626741339097 ✓ + 85.942 ppm	8.626741339097 ✓ + 85.942 ppm
GR perihelion precession (Earth) (arcsec/century)	3.840000000000 ±13020.833 ppm	3.839649170448 ✓ − 91.362 ppm	3.839649170448 ✓ − 91.362 ppm	3.839649170448 ✓ − 91.362 ppm
GR perihelion precession (Mars) (arcsec/century)	1.350000000000 ±37037.037 ppm	1.351262711937 ✓ + 935.342 ppm	1.351262711937 ✓ + 935.342 ppm	1.351262711937 ✓ + 935.342 ppm
GR perihelion precession (Jupiter) (arcsec/century)	0.06230000000000 ±16051.364 ppm	0.06232950659433 ✓ + 473.621 ppm	0.06232950659433 ✓ + 473.621 ppm	0.06232950659433 ✓ + 473.621 ppm
Light deflection (solar limb) (arcsec)	1.751700000000 ±456.699 ppm	1.751644036032 ✓ − 31.948 ppm	1.751644036032 ✓ − 31.948 ppm	1.751644036032 ✓ − 31.948 ppm
Shapiro round-trip (Earth-Saturn conj., $b = 1.6\,R_\odot$) (μs)	260.0000000000 ±76923.077 ppm	264.4693472015 ✓ + 17189.797 ppm	264.4693472015 ✓ + 17189.797 ppm	264.4693472015 ✓ + 17189.797 ppm
Solar grav. redshift z (limb)	2.120000000000e-6 ±4716.981 ppm	2.123052482755e-6 ✓ + 1439.850 ppm	2.123052482755e-6 ✓ + 1439.850 ppm	2.123052482755e-6 ✓ + 1439.850 ppm
Schwarzschild radius (Sun) (km)	2.953000000000 ±338.639 ppm	2.954015224506 • + 343.794 ppm	2.954015224506 • + 343.794 ppm	2.954015224506 • + 343.794 ppm
Schwarzschild radius (Earth) (mm)	8.870000000000 ±112.740 ppm	8.870125356371 ✓ + 14.133 ppm	8.870125356371 ✓ + 14.133 ppm	8.870125356371 ✓ + 14.133 ppm
MOND scale $a_0 = cH_0/(2\pi)$ ($\mathrm{m/s^2}$)	1.2×10^{-10} ±0.02 stat ±0.24 syst	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open
$\Lambda_{\mathrm{crit-tension}} = (H_0/c)^2$ ($\mathrm{m^{-2}}$)	1.097121526139e-52	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open
Hubble constant H_0 (km/s/Mpc)	$\sim$ 73 (local) $\sim$ 67 (CMB)	open †† regime tension; pending	open †† regime tension; pending	open †† regime tension; pending

IV. PREDICTED BUT UNOBSERVED (FALSIFICATION TARGETS)

Discrete dark-matter spectrum (21 modes total: 13 with specific mass predictions, 8 high-mass with envelope). The framework predicts a discrete spectrum of dark-matter masses, none yet detected. The 13 specific-mass predictions are:

- *Ultralight band (7 modes):* 0.62 μeV, 2.23 neV, 8.03 peV, 28.9 feV, 0.10 aeV, 0.37 zeV, 1.34 zeV. The Lyman-α-forest ultralight-DM bound $m \gtrsim 2\times10^{-20}$ eV at 95% CL [10] excludes the 1.34 zeV mode; the next-lightest mode (0.37 zeV is below; 0.10 aeV $\sim$ 19$\times$ above the bound) is above the bound.
- *MeV–GeV band (6 modes):* 0.83 MeV, 5.83 MeV, 46.6 MeV, 1.49 GeV, 4.82 GeV, 593 GeV.

The remaining 8 high-mass modes lie within the envelope [191 GeV, 1.1 EeV]; specific masses ‡ open.

CP rationals (framework predictions). $\delta_{\rm CP}^{\rm PMNS} = 11\pi/10 \approx 198^\circ$ (consistent with NuFit-6.0 [2] normal ordering: at the IC24 fit $\delta_{\rm CP}^{\rm best} = 212^\circ$ with asymmetric range $-41^\circ/+26^\circ$, the framework's 198° sits at $\sim 0.3\sigma$ on the low-δ side; awaiting DUNE/HyperK). $\delta_{\rm CKM} = 11\pi/30 \approx 1.152$ rad (within $\sim 1\sigma$ of the CKMfitter/PDG 2024 central value [1, 7]).

Three matter generations ($N_{\rm gen} = 3$). The framework predicts exactly three chiral fermion generations.

Strong CP: $\theta_{\rm QCD} = 0$ exactly, with no PQ axion required.

Internal topology-discrimination test: $|t| = 6.92\sigma$. A measurement-level discrimination test against the leading alternative scenario clears 5σ on both percentile and BCa 95%-CI floors.

No quantum-gravity scale below $\sim 10\,E_{\rm Planck}$; no fourth chiral generation; no fundamental scalar beyond Higgs.

V. WHAT WOULD FALSIFY THE FRAMEWORK

The list below is mixed: items marked † are committed and falsifiable today; items marked ‡ become falsifiable once upstream prediction work closes (see § Work remaining).

† Fourth chiral fermion generation observed.

† $\theta_{\rm QCD}$ measured nonzero.

‡ Direct-detection dark-matter peak at a mass either (a) outside the union of the 13 specific-mass predictions ± predicted width AND outside the high-mass envelope [191 GeV, 1.1 EeV], or (b) inside the high-mass envelope but inconsistent with the 8 high-mass values once those are committed.

‡ $\delta_{\rm CP}^{\rm PMNS}$ measured inconsistent with $11\pi/10$ at $\geq 3\sigma$ in normal ordering. (Currently auto-flagged blue; becomes a committed-falsifiable prediction once the upstream PMNS-mixing work closes.)

† Topology-discrimination null result: a reproducible measurement-level test showing the framework's preferred scenario loses to the leading alternative at $> 5\sigma$.

† Astrophysical detection of an azimuthal mode-pattern on a rotating compact body (e.g. pulsar surface oscillations via NICER or successor X-ray timing missions) whose symmetry group, classified as a finite subgroup of $O(3)$, falls outside $\{\mathbb{Z}_n, D_n, T, O, I\}$.

‡ Stable non-collapsed compact object whose mass exceeds the framework-predicted substrate TOV bound $M_{\rm sub-TOV}$ (value ‡ open; bound is geometric at saturation, $r_{\rm phys} = r_s$).

† CMB-S4 measurement of n_s inconsistent with the framework's prediction $n_s = (N-2)/N$ at the $\Delta n_s \sim 10^{-4}$ level, with $N = 60$ as the framework canonical value ($n_s = 58/60 = 0.9667$ at $N = 60$ vs. $n_s = 56/58 = 0.9655$ at $N = 58$ differ by $\sim 1.2\times10^{-3}$, so the N-dependence is itself part of the falsifiable claim).

† Frame-dependent c, vacuum dispersion, or any departure from speed-of-light universality at any energy scale.

† Experimental confirmation of a quantum-gravity scale below $\sim 10\,E_{\rm Planck} \approx 10^{20}$ GeV (via gravitational decoherence at observer-energies $\ll E_{\rm Planck}$, or any single-graviton-exchange process detected at sub-Planckian energies).

VI. WORK REMAINING (PRIORITIZED BY UNBLOCK COUNT)

The two tables together carry 38 distinct blue-flagged observables that are *not yet falsifiable* because the upstream prediction work that would commit them at structural precision is not yet closed. The list below tallies open work items in priority order, counting how many blue observables each item turns into committed predictions.

a. J1. Lepton/quark mixing structure — unblocks 12 observables. Blocks: PMNS $\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, $\delta_{\rm CP}$ (4); CKM $\sin^2\theta_{12}^{\rm CKM}$, $\sin^2\theta_{23}^{\rm CKM}$, $\sin^2\theta_{13}^{\rm CKM}$, $\delta_{\rm CKM}$ (4); Higgs partial widths $\Gamma_{H\to bb,\tau\tau,cc,ff}$ (4).

b. J2. Gauge-coupling precision — unblocks 9 observables. Blocks: $\alpha_{\rm EM}^{\rm tree}$, α_2, $\alpha_1^{\rm EW}$, $\alpha_1^{\rm GUT}$ (4); g_2, $g_1^{\rm EW}$, $g_1^{\rm GUT}$ (3); $\sin^2\theta_W$ (1); Z boson cascading from the W-mass+$\sin^2\theta_W$ identity (1).

c. J3. $\alpha_{\rm EM}$ hadronic running — unblocks 5 observables. Blocks: lepton anomalous magnetic moments $a_e^{\rm LO}$, $a_\mu^{\rm LO}$, $a_\tau^{\rm LO}$ (3); Bohr radius a_0 (1); Rydberg energy Ry (1).

d. J4. Gravity / cosmology block — unblocks 4 observables. A single closure decomposes into five sub-items: (a) the form of the MOND interpolation function (cf. Famaey–Binney [4] and the radial-acceleration relation [5]); (b) the prefactor in the relation $a_0 = cH_0/(2\pi)$ (cf. Milgrom 1983 [6]); (c) the residual $\sim 50\%$ Friedmann coefficient that closes $\Lambda_{\rm crit-tension} = (H_0/c)^2$ to the full ΛCDM relation; (d) the local-vs-CMB H_0 regime tension

at $\sim 9\%$ (SH0ES [9] vs. Planck [8]); (e) the dark-mode spatial density at cluster scales needed for combined-MOND-plus-dark-mode acceleration tests. Blocks: ρ_Λ (dark-tower contribution); MOND a_0; $\Lambda_{\mathrm{crit-tension}}$; H_0 regime tension.

e. J5. NLO electroweak radiative corrections — unblocks 4 observables. Blocks: $\Gamma_W, \Gamma_Z, \Gamma_Z^{\mathrm{invisible}}, \Gamma_Z^{\mathrm{leptonic}}$ (4).

f. J6. NLO+ QED/QCD corrections for lepton lifetimes — unblocks 2 observables. Blocks: τ_μ, τ_τ (2).

g. J7. Λ_{QCD} *closure — unblocks 1 observable.* Blocks: α_s at lower scales (1).

h. J8. Thermal-history closure — unblocks 1 observable. Blocks: N_{eff} at recombination (1).

i. Total unblock count. J1 (12) + J2 (9) + J3 (5) + J4 (4) + J5 (4) + J6 (2) + J7 (1) + J8 (1) = 38 observables, matching the count of distinct blue-flagged observables across both tables.

j. Strategic note. J1 and J2 together account for 21 of 38 blue observables (55%); J3 and J7 are tightly coupled. Realistically, closing J1+J2+J3+J7 together unblocks 27 of 38 (71%); J4 is structurally independent and is the second-priority closure.

k. ‡*-flagged items.* ρ_Λ and the H_0 regime tension fall under J4 (sub-items c and d). The TOV bound's nuclear-scale closure is a separate item not yet in the J1–J8 tally; it remains ‡-flagged.

VII. PROVENANCE

All numerical content auto-generated at mp.prec = 500. Source code, per-section reproducibility traces, and JSON output artefacts mirrored at github.com/gable-digital/papers.

[1] J. Charles *et al.* (CKMfitter Group), "CP violation and the CKM matrix: assessing the impact of the asymmetric B factories," *Eur. Phys. J. C* **41**, 1–131 (2005), doi:10.1140/epjc/s2005-02169-1, arXiv:hep-ph/0406184.

[2] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, "NuFit-6.0: updated global analysis of three-flavor neutrino oscillations," *J. High Energy Phys.* **2024**, 216 (2024), doi:10.1007/JHEP12(2024)216, arXiv:2410.05380.

[3] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, "CODATA recommended values of the fundamental physical constants: 2022," *Rev. Mod. Phys.* **97**, 025002 (2025), doi:10.1103/RevModPhys.97.025002, arXiv:2409.03787.

[4] B. Famaey and J. Binney, "Modified Newtonian dynamics in the Milky Way," *Mon. Not. R. Astron. Soc.* **363**, 603–608 (2005), doi:10.1111/j.1365-2966.2005.09474.x, arXiv:astro-ph/0506723.

[5] S. S. McGaugh, F. Lelli, and J. M. Schombert, "The radial acceleration relation in rotationally supported galaxies," *Phys. Rev. Lett.* **117**, 201101 (2016), doi:10.1103/PhysRevLett.117.201101, arXiv:1609.05917.

[6] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," *Astrophys. J.* **270**, 365–370 (1983), doi:10.1086/161130.

[7] S. Navas *et al.* (Particle Data Group), "Review of particle physics," *Phys. Rev. D* **110**, 030001 (2024), doi:10.1103/PhysRevD.110.030001.

[8] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020), doi:10.1051/0004-6361/201833910, arXiv:1807.06209.

[9] A. G. Riess *et al.*, "A comprehensive measurement of the local value of the Hubble constant with 1 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ uncertainty from the Hubble Space Telescope and the SH0ES Team," *Astrophys. J. Lett.* **934**, L7 (2022), doi:10.3847/2041-8213/ac5c5b, arXiv:2112.04510.

[10] K. K. Rogers and H. V. Peiris, "Strong bound on canonical ultra-light axion dark matter from the Lyman-alpha forest," *Phys. Rev. Lett.* **126**, 071302 (2021), doi:10.1103/PhysRevLett.126.071302, arXiv:2007.12705.